

Elijah Ballou

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PROFILE

Computer science student, Unity VR developer, and published undergraduate researcher focused on human-computer interaction, immersive technology, and accessible systems. Experience developing XR simulations, educational games, and research prototypes using Unity, C#, and Python.

TECHNICAL SKILLS

- **Programming:** C#, Python, JavaScript
- **XR & Game Development:** Unity, OpenXR, XR Interaction Toolkit, Meta Quest
- **Web & Data:** HTML, CSS, React, Node.js, Flask, MongoDB
- **Tools & Design:** Git, GitHub, VS Code, Blender, Figma, Canva
- **Research:** Human-Computer Interaction, Brain-Computer Interfaces, Interactive Systems

EXPERIENCE

Noblis May–Aug 2026
Unity VR Developer Intern Reston, VA

- Developed and tested Unity/C# VR prototypes with XR interactions, UI, and Meta Quest gameplay; debugged for usability and performance.

Human-AI Experience (HAX) Lab Jan 2023–Present
Undergraduate Research Assistant Baltimore, MD

- Develop VR, AI, and intelligent tutoring research prototypes; design HCI-informed interfaces and contribute literature reviews.

Introduction to Game Design and Development Aug 2023–Present
Undergraduate Teaching Assistant Baltimore, MD

- Lead Unity labs and support students with C# scripting, debugging, and game development.
- Manage course resources and assess projects using established grading criteria.

SELECTED PROJECTS

ORBIT XR: Extended Reality Training Simulation 2026–Present
Lead Developer | Unity, C#, XR, Meta Quest
• Design XR situational-awareness training scenarios for police officers and academy recruits; plan workload and usability evaluations.

Mind Mastery: Brain-Computer Interface Gaming 2025
Co-Lead Developer | Unity, C#, Python, Muse 2, EEG
• Co-developed an accessible hands-free game integrating Muse 2, Python, Unity, and EEG-based player controls.

Interactive Python Learning Platform Feb 2025–Present
Team Lead | Python, PyTorch, MongoDB, HTML/CSS, Unity, C#
• Develop a gamified adaptive tutoring platform with text-to-speech, captions, and customizable interfaces.

EDUCATION

Morgan State University Baltimore, MD
B.S. Computer Science Jan 2023–Present

GPA: 3.333/4.000

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PUBLICATIONS & RECOGNITION

PUBLICATIONS

- [1] Odunze, O., **Ballou, E.**, Kanu, A., Kelly, A., Void, E., & Mack, N. (2025).
Inclusive Gaming Through Brain-Computer Interfaces: The Mind Mastery Experience .
Human-Computer Interaction & Emerging Technologies, AHFE, vol. 195.
- [2] **Ballou, E.**, Odunze, O., Adeleke, M., & Mack, N. (2025).
PyPro: Think in Code. Grow in Logic!
Human-Computer Interaction & Emerging Technologies, AHFE, vol. 195.
- [3] Mack, N. A., Tandjong, C. W., Adeleke, M. B., **Ballou, E.**, Harry, A., & Panton, J. (2025).
Revolutionizing Culturally Relevant Math Education Through AI and Co-Designing Strategies .
CHI Extended Abstracts.
- [4] Mack, N. A., Adeleke, M. B., **Ballou, E.**, Davis, D., Ingram, V., & Cox, K. (2024).
Breaking Stereotypes and Feeding the STEM Pipeline .
SIGCSE 2024, vol. 1, pp. 771–777.
- [5] Mack, N. A., Adeleke, M. B., Ingram, V., **Ballou, E.**, Briggs-Belt, J. K., & Jordan, A. (2024).
CodeBears: Key Insights Gained from a Summer Coding Camp Empowering Underrepresented Youth .
BICE 2024, pp. 80–86.
- [6] Meekins, S., **Ballou, E.**, & Mack, N. A. (2024).
Dreadphobia: Evaluating the Usability of a Virtual Reality Application in Support of Mental Health .
HCII 2023 Late Breaking Posters, CCIS vol. 1957, Springer.

HONORS & AWARDS

- **1st Place, Poster Presentation** — Morgan TechFest, Morgan State University (2023)
- **2nd Place, Poster Presentation** — CMD-IT/ACM Richard Tapia Conference; Tapia Scholarship Recipient (2023)
- **2nd Place** — Morgan State CodeLinc Hackathon (2023)
- **Most Technically Challenging** — HBCU Game Jam, Spelman College (2023)

PROFESSIONAL MEMBERSHIPS

Alpha Sigma Lambda	2025–Present
National Society of Black Engineers	2024–Present
Association for Computing Machinery	2023–Present